

WISER: Wider Search, Deeper Thinking, and Adaptive Fusion for Training-Free Zero-Shot Composed Image Retrieval

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Abstract

*Zero-Shot Composed Image Retrieval (ZS-CIR) aims to retrieve target images given a multimodal query (comprising a reference image and a modification text), without training on annotated triplets. Existing methods typically convert the multimodal query into a single modality—either as an edited caption for Text-to-Image retrieval (T2I) or as an edited image for Image-to-Image retrieval (I2I). However, T2I often loses fine-grained visual details, while I2I struggles with complex semantic modifications. To effectively leverage their complementary strengths under diverse query intents, we propose **WISER**, a training-free framework that unifies T2I and I2I via a “retrieve–verify–refine” pipeline, explicitly modeling intent awareness and uncertainty awareness. Specifically, WISER first performs **Wider Search** by generating both edited captions and images for parallel retrieval to broaden the candidate pool. Then, it conducts **Adaptive Fusion** with a verifier to assess retrieval confidence, triggering refinement for uncertain retrievals, and dynamically fusing the dual-path for reliable ones. For uncertain retrievals, WISER generates refinement suggestions through structured self-reflection to guide the next retrieval round toward **Deeper Thinking**. Extensive experiments show the superior performance of WISER across multiple benchmarks, achieving relative improvements of 45% on CIRCO (mAP@5) and 57% on CIRR (Recall@1) over existing training-free methods. Code is released at <https://github.com/Physicsmile/WISER>.*

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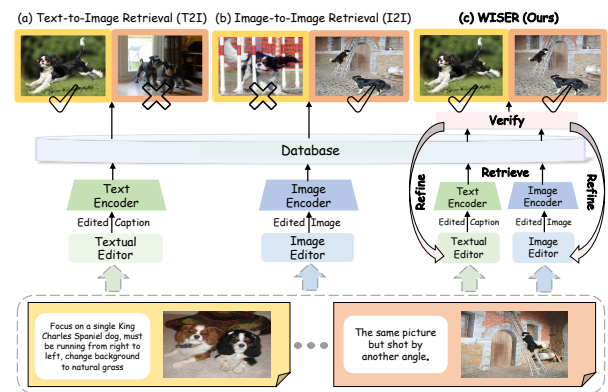


Figure 1. **Comparison of existing ZS-CIR methods.** (a) T2I may fail to preserve visual details from the reference image, while (b) I2I often struggles with complex modifications. In contrast, (c) WISER successfully adapts to diverse modification intents through a “retrieve–verify–refine” pipeline.

1. Introduction

Imagine you aim to find a red leather jacket similar to the one your friend is wearing, but with a hood instead of a collar. You show the jacket to a retrieval system and say, “Add a hood.” This scenario conveys the essence of Composed Image Retrieval (CIR): retrieving a target image that matches a composed query consisting of a reference image and a modification text [4, 12, 23, 24, 27, 30, 39, 51, 55]. CIR plays a vital role in applications such as fashion search and product recommendation.

Most existing CIR methods [5, 17, 22, 38, 43, 51, 54] rely on costly annotated triplets, which are labor-intensive and difficult to scale to new domains. To alleviate this de-

pendency, Zero-Shot CIR (ZS-CIR) has recently been introduced [6, 48], mainly following two paradigms, as shown in Figure 1. The first paradigm leverages the text editor (usually implemented with a textual inversion module [6, 21, 48] or an image captioner cascaded with a Large Language Model [15, 28, 58]) to transform the composed query into an edited caption, and then performs Text-to-Image retrieval (T2I). Although this paradigm shows promise for complex semantic modifications, it often fails to preserve fine-grained visual details from the reference image. The second paradigm, in contrast, employs an image editor to produce an edited image by editing the reference image conditioned on the modification text, thus framing CIR as Image-to-Image retrieval (I2I) [20, 33, 52]. It retains visual details better but performs poorly when the query intent is ambiguous or involves complex compositional edits. Given the diversity of real-world CIR queries, relying solely on either T2I or I2I is insufficient. This raises a crucial question: *How can we leverage the complementary strengths of both paradigms to accommodate diverse modification intents?* However, effectively unifying T2I and I2I for ZS-CIR is non-trivial, due to the following two key challenges: (1) *Intent Awareness*. Existing methods often adopt static (e.g., fixed-weight) fusion strategies [33, 52], which lack adaptability to varying query intents. (2) *Uncertainty Awareness*. Current approaches overlook the uncertainty of candidates from each branch, leading to unreliable fusion.

To address these challenges, we propose **WISER**, a training-free framework to enable Wider Search, Adaptive Fusion, and Deeper Thinking for ZS-CIR. WISER generalizes well across domains without additional training, and its modular design is compatible with off-the-shelf models. Specifically, WISER unifies T2I and I2I paradigms through a “retrieve–verify–refine” pipeline, explicitly modeling intent and uncertainty awareness to handle diverse modification intents. Firstly, we perform *Wider Search* by activating both T2I and I2I pathways in parallel. It uses an editor to produce an edited caption and an edited image, expanding the candidate pool from two complementary views. Next, we conduct *Adaptive Fusion* by employing a verifier to evaluate candidates from each branch with confidence scores, determining the reliability of each pathway. For uncertain retrievals, it triggers a refinement process; otherwise, it applies a multi-level fusion strategy that addresses uncertainty awareness at the branch level and intent awareness at the candidate level. Finally, we enable *Deeper Thinking* by refining uncertain retrievals via a refiner, which generates improvement suggestions through structured self-reflection to guide the next retrieval round. The pipeline iterates until a maximum count is reached.

In summary, our main contributions are as follows: (1) We are the first to propose a training-free framework for ZS-CIR that adaptively leverages the complementary

strengths of both T2I and I2I paradigms. (2) WISER unifies T2I and I2I through an iterative “retrieve–verify–refine” loop, enabling both intent and uncertainty awareness. (3) WISER demonstrates remarkable superiority and generalization across multiple benchmarks, even surpassing many training-based methods.

2. Related Work

2.1. Zero-Shot Composed Image Retrieval

Composed Image Retrieval (CIR) retrieves target images given a reference image and a modification text as the query. Supervised methods [5, 17, 22, 38, 51, 54] rely on costly annotated triplets, while Zero-Shot CIR (ZS-CIR) leverages the intrinsic generalization of pretrained vision-language models without depending on any manual triplet annotation [6, 16, 48]. Existing ZS-CIR methods mainly follow two paradigms. Text-to-Image retrieval (T2I) approaches [6, 21, 28, 48, 49, 58], such as Pic2Word [48] and SEARLE [6], need training-dependent textual inversion modules and suffer from limited visual expressiveness. Training-free methods such as CIReVL [28] and CoTMR [49] utilize Large Language Models (LLMs) or Multimodal Large Language Models (MLLMs) to infer target captions through multimodal reasoning, excelling at semantic changes but potentially losing visual details. Image-to-Image retrieval (I2I) approaches, exemplified by CompoDiff [20], employ diffusion-based frameworks in CLIP [46] feature space to preserve visual details, though they may struggle with complex edits. Recent methods, such as CIG [52] and IP-CIR [33] edit images via diffusion models [42, 47, 62] to augment T2I baselines, yet they still depend on additional training or the manual tuning of fusion hyperparameters. In contrast, our work unifies both paradigms in a training-free framework that dynamically balances T2I and I2I, overcoming limitations of static fusion strategies.

2.2. Vision-Language Models for CIR

Vision-language models (VLMs) learn aligned representations from large-scale image-text pairs [31, 32, 46, 59], mapping visual and textual inputs into a shared embedding space [11, 44, 45, 61]. The CLIP model [46] has demonstrated remarkable zero-shot capabilities across diverse applications. Beyond CIR, VLMs have shown strong performance in zero-shot image classification [10, 29, 36, 46], visual question answering (VQA) [9, 40, 60], semantic segmentation [8], recommendation systems [19, 25], and social network analysis [13, 14, 63]. Most ZS-CIR methods rely on CLIP variants, which either convert the multimodal query into text embedding space as T2I or into image embedding space as I2I. However, they fail to adaptively harness both strengths based on the specific modification intent. Our framework overcomes this limitation by adap-

tively integrating both T2I and I2I pathways through a novel “retrieve–verify–refine” pipeline.

3. Method

In this section, we present **WISER**, a training-free framework for ZS-CIR. As illustrated in Figure 2, WISER employs a “retrieve–verify–refine” pipeline that unifies T2I and I2I pathways through three core components: Wider Search enables dual-path retrieval (Sec. 3.1), Adaptive Fusion dynamically evaluates and fuses candidates based on confidence scores (Sec. 3.2), and Deeper Thinking refines uncertain retrievals (Sec. 3.3). Formally, given a composed query including a reference image I_{ref} and a modification text T_{mod} , the goal of CIR is to retrieve target images from a database $\mathcal{D}$ to match the composed query. In our work, we first employ an *editor* $\mathcal{F}$ to derive an edited image I_{edit} and an edited caption C_{edit} . They are then encoded into query vectors q_v and q_t using a visual encoder E_{img} and a text encoder E_{txt} [46], respectively. The final retrieval is performed by computing the cosine similarity between these query vectors and the candidates in $\mathcal{D}$.

3.1. Wider Search: Dual-path Retrieval

To handle diverse retrieval needs in ZS-CIR, we design *Wider Search*, a strategy that activates both T2I and I2I in parallel to broaden the candidate pool.

Text-to-Image Retrieval (T2I). We first use a pre-trained captioner to encode the reference image I_{ref} into a descriptive caption C_{ref} . Subsequently, the editor integrates C_{ref} with the modification text T_{mod} through its understanding capability, generating an edited caption C_{edit} that explicitly describes the target image. Formally:

$$C_{\text{edit}} = \mathcal{F}_{\text{txt}}(C_{\text{ref}}, T_{\text{mod}}), \quad (1)$$

where $\mathcal{F}_{\text{txt}}$ denotes the textual editing function of the editor. This process aims to capture complex semantic modifications specified in the text.

Image-to-Image Retrieval (I2I). In parallel, the editor leverages its image editing capability to generate an edited image I_{edit} directly from I_{ref} and T_{mod} :

$$I_{\text{edit}} = \mathcal{F}_{\text{img}}(I_{\text{ref}}, T_{\text{mod}}), \quad (2)$$

where $\mathcal{F}_{\text{img}}$ denotes the image editing function. I_{edit} retains detailed visual attributes (e.g., texture, style) of the reference image while applying the required modifications.

Dual-path Retrieval. Given the edited caption C_{edit} and the edited image I_{edit} , we perform CLIP-based retrieval through two pathways. For each pathway $p \in \{\text{T2I}, \text{I2I}\}$, we retrieve the top- K candidate images:

$$\mathcal{R}_p = \{I_p^1, I_p^2, \dots, I_p^K\}. \quad (3)$$

We then take the union of both candidate sets to form an expanded candidate pool:

$$\mathcal{R}_{\text{union}} = \mathcal{R}_{\text{T2I}} \cup \mathcal{R}_{\text{I2I}}. \quad (4)$$

This operation ensures that all potentially relevant candidates from both T2I and I2I retrievals are considered, thereby improving retrieval recall.

3.2. Adaptive Fusion: Verification-guided Dual-path Integration

While *Wider Search* expands the retrieval space by independently producing semantically and visually aligned candidates, a simple union of the dual-path fails to dynamically balance their complementary strengths under diverse query intents. To address this, we introduce *Adaptive Fusion*, a verification-guided module that explicitly models uncertainty and intent for effective fusion. It first verifies candidates to assess the reliability of each pathway. If the retrieval is deemed uncertain, it triggers *Deeper Thinking* (Sec. 3.3) for refinement; otherwise, it dynamically integrates candidates from both paths via a multi-level fusion strategy.

Verification-based Scoring. For each candidate $I_p^k \in \mathcal{R}_{\text{union}}$, we construct a triplet $(I_{\text{ref}}, T_{\text{mod}}, I_p^k)$ and evaluate it using a VLM-based *verifier* Φ to assess whether the modification intent is faithfully reflected in the candidate image. The verifier is asked to answer a binary question: “*Decide if the candidate image matches the result of applying the instruction to the reference image.*” Let $\ell_{p,k}^{(\text{yes})}$ and $\ell_{p,k}^{(\text{no})}$ denote the logits corresponding to the answer of “yes” and “no”, respectively:

$$\{\ell_{p,k}^{(\text{yes})}, \ell_{p,k}^{(\text{no})}\} = \Phi(I_{\text{ref}}, T_{\text{mod}}, I_p^k). \quad (5)$$

The confidence score c_p^k that I_p^k is a correct modification is computed as:

$$c_p^k = \frac{\exp(\ell_{p,k}^{(\text{yes})})}{\exp(\ell_{p,k}^{(\text{yes})}) + \exp(\ell_{p,k}^{(\text{no})})}. \quad (6)$$

A higher confidence score indicates stronger alignment between the candidate and the query intent.

Multi-Level Fusion Strategy. Our fusion strategy operates at two levels to address uncertainty and intent awareness, respectively.

Branch-Level Uncertainty Awareness. We first assess the reliability of each pathway by identifying the most promising candidate from each pathway as the pseudo-target I_p^* :

$$I_p^* = \arg \max_k c_p^k, \quad r_p = \max_k c_p^k, \quad (7)$$

where r_p represents the reliability score of pathway p . If $\min(r_{\text{T2I}}, r_{\text{I2I}}) < \tau$ (i.e., either pathway demonstrates uncertainty), the corresponding pseudo-target is forwarded to

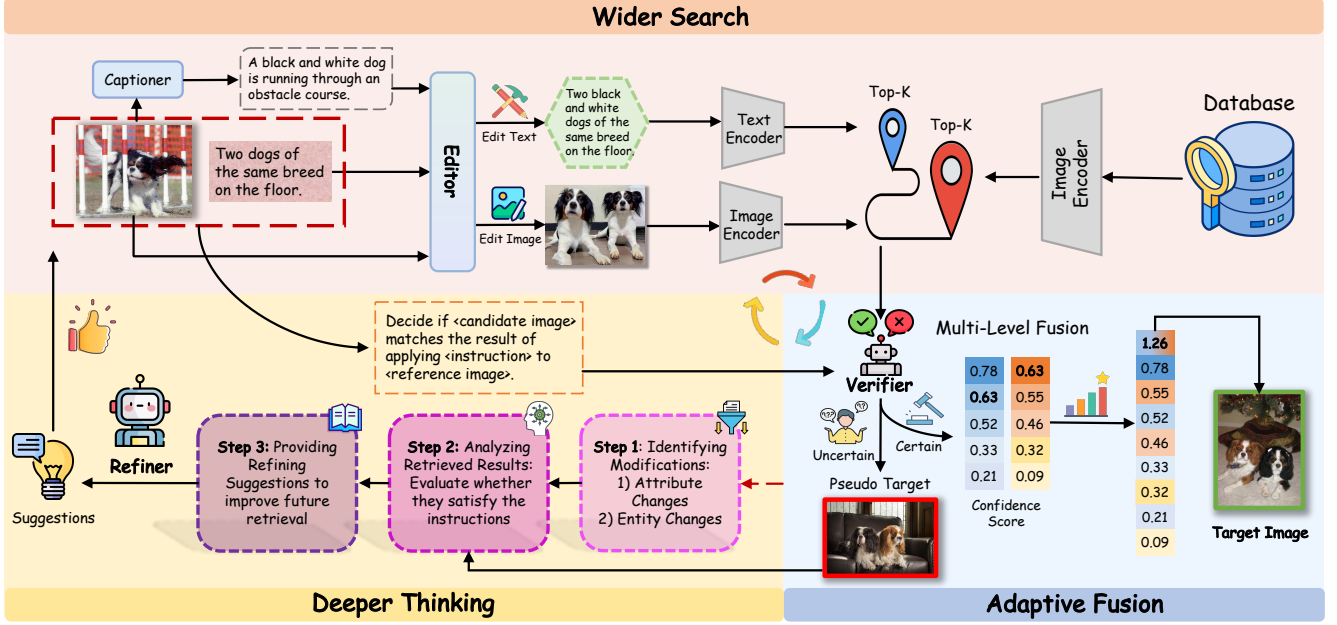


Figure 2. **Overview of the proposed WISER framework.** (1) Wider Search. We leverage an editor to produce text and image queries for dual-path retrieval, aggregating the top- K results into a unified candidate pool. (2) Adaptive Fusion. We employ a verifier to assess the candidates with confidence scores, applying a multi-level fusion strategy for high-confidence results and triggering refinement for low-confidence ones. (3) Deeper Thinking. For uncertain retrievals, we leverage a refiner to analyze unmet modifications and then feed targeted suggestions back to the editor, iterating until a predefined limit is reached.

Deeper Thinking (Sec. 3.3) for refinement rather than proceeding to fusion.

Candidate-Level Intent Awareness. For reliable retrievals, we dynamically integrate candidates through intent-aware fusion. We first compute a fused confidence score that aggregates evidence from both pathways:

$$c_{\text{fused}}^k = c_{\text{T2I}}^k + c_{\text{I2I}}^k. \quad (8)$$

T2I typically yields higher confidence for semantic-heavy edits, while I2I excels when visual details are key. When both aspects matter, candidates strong in both semantics and visuals gain higher fused scores and superior ranks. Then, we sort $\mathcal{R}_{\text{union}}$ using a lexicographical order defined by:

$$\Psi(I^k) = (-c_{\text{fused}}^k, -\max(c_{\text{T2I}}^k, c_{\text{I2I}}^k), -c_{\text{T2I}}^k). \quad (9)$$

The primary sorting key is the fused score, which captures overall intent alignment. For tie-breaking, it uses maximum single-path confidence and T2I confidence to provide nuanced, intent-aware disambiguation. Unlike conventional fusion strategies that rely on fixed weights or additional training modules, we dynamically integrate the two retrieval pathways with both intent and uncertainty awareness.

3.3. Deeper Thinking: Structured Self-reflection for Uncertain Retrievals

For uncertain retrievals from Adaptive Fusion ($r_p < \tau$), we invoke *Deeper Thinking*, a refinement module that im-

proves the quality of the corresponding edited caption and edited image by analyzing modification failures and generating targeted suggestions. The refinement is driven by an LLM-based *refiner* that performs a three-step analysis:

Step 1: Identifying Modifications. Given the caption of the reference image C_{ref} and the modification text T_{mod} , the refiner deeply analyzes the intended changes and generates structured modification phrases. Specifically, it identifies two types of modifications: (1) Attribute Changes: If the modification involves changing characteristics of an entity in I_{ref} , the refiner specifies the change; (2) Entity Additions/Deletions: If the modification adds or removes an entity, the refiner specifies the operation.

Step 2: Analyzing Retrieved Results. We first obtain the caption of the pseudo-target I_p^* . It is then compared with the modification phrases from Step 1 to determine whether the retrieved image satisfies the user’s instruction. This comparison reveals which aspects of the modification are missed or incorrectly applied.

Step 3: Providing Refining Suggestions. For any unmet modifications, the refiner proposes concise, targeted suggestions to improve future retrieval: For T2I, it generates a textual suggestion to enhance the edited caption. For I2I, it provides visual guidance to improve the edited image generation. Otherwise, the refinement loop is terminated, and the current retrieved images are kept as the final output.

The suggestions are then concatenated with the modification text and fed back to the editor $\mathcal{F}$ to regenerate refined C_{edit} or I_{edit} , continuing the “retrieve–verify–refine” cycle. This process iterates until the maximum number of iterations N is reached. Deeper Thinking mimics human-like introspection, enabling WISER to self-reflect and adapt to complex or ambiguous queries without any training.

4. Experiments

We conduct extensive experiments to investigate the effectiveness of our proposed method.

4.1. Experimental Setup

Datasets. We evaluate our proposed WISER on three ZS-CIR benchmarks: **Fashion-IQ** [57], **CIRR** [38] and **CIRCO** [6]. Fashion-IQ focuses on fashion retrieval with three subcategories (Dress, Shirt, Tootie). We use Recall@10, Recall@50 as the evaluation metric, and their mean across the three categories. CIRR is the first natural image dataset designed specifically for CIR, built from real-life images in an open domain. We report Recall@ k ($k \in \{1, 5, 10, 50\}$) and Recall_{Subset}@ k ($k \in \{1, 2, 3\}$) following the benchmark protocol. CIRCO is built from the COCO 2017 unlabeled set and is the first CIR dataset providing multiple ground truths per query. It contains 123,403 images in the database and 800 test queries. Following prior works, we report mAP@ k ($k \in \{5, 10, 25, 50\}$) as the evaluation metric.

Implementation Details. We adopt BAGEL [18] as the editor, Qwen2.5-VL-7b [3] as the verifier, and GPT-4o [1] as the refiner. Following previous methods [28], we use the pre-trained BLIP-2 [32] as our captioner. For the retrieval model, we experiment with different CLIP variants, including ViT-B/32, ViT-L/14, and ViT-G/14 CLIP from OpenCLIP [26]. The retrieval candidate pool size K is set to 50, and the reliability threshold τ is 0.7. For uncertain cases, the refinement iteration is set to one round by default. All experiments are implemented using PyTorch [41] on a single NVIDIA H20 GPU, and the results are reported on official validation or test splits.

Baselines. We compare WISER with a wide range of representative ZS-CIR baselines. Specifically, PALAVRA [16], SEARLE [6], LinCIR [21], MOA [35] and HIT [34] are textual inversion approaches originally designed or adapted for the ZS-CIR task. We also include recent training-free methods such as CIReVL [28], LDRE [58], OSrCIR [50], AutoCIR [15], and CoTMR [49], which leverage LLMs or MLLMs without additional training. We also compare with IP-CIR [33] under the training-free setting reported in paper, which fuses T2I and I2I pathways using a fixed weight.

4.2. ZS-CIR Benchmark Comparisons

We conduct comprehensive comparisons between WISER and state-of-the-art ZS-CIR methods on three benchmark datasets: CIRCO, CIRR, and Fashion-IQ. The results are summarized in Table 1 and Table 2.

CIRCO. As shown in the left section of Table 1, WISER achieves strong performance on the CIRCO dataset. Key observations are as follows: (1) WISER outperforms both training-free and training-based methods across all CLIP backbones. It achieves a relative improvement of 44.98% in mAP@5 over CoTMR with ViT-B/32, and surpasses LinCIR by 22.51% in mAP@5 with ViT-L/14. (2) WISER consistently achieves superior mAP across all k by integrating candidates from both T2I and I2I pathways, expanding the search space and improving retrieval under the multi-target nature of CIRCO.

CIRR. The right section of Table 1 presents results on the challenging CIRR dataset. Key observations are as follows: (1) WISER delivers strong performance. With ViT-B/32, it achieves a relative improvement of 56.98% in Recall@1 over the best baseline, demonstrating robust capability in handling diverse and ambiguous modification intents. (2) In subset recall evaluation, which requires retrieving the correct image from six curated samples, WISER achieves 77.30% Recall_{sub}@1 with ViT-B/32, surpassing the second-best method by 10.69%. The improvement is more pronounced for smaller k , confirming its effectiveness in ranking the most relevant images at the top. (3) While some baselines show performance saturation with larger backbones, WISER exhibits consistent gains, with ViT-G/14 achieving a Recall@1 of 49.54%, far exceeding other methods. This scalability validates the generalizability of our framework.

Fashion-IQ. Table 2 reports results on the Fashion-IQ validation set, which focuses on fine-grained attribute modifications in the fashion domain. Key observations are as follows: (1) WISER outperforms both training-free and training-dependent baselines using ViT-B/32 and ViT-L/14 by leveraging complementary information from the T2I and I2I pathways, balancing semantic accuracy with visual consistency. (2) With ViT-G/14, WISER achieves comparable or superior performance to training-based methods such as LinCIR despite lacking their training advantage, further confirming the strong generalization and effectiveness of our approach.

4.3. Ablation Study

In this section, we conduct experiments on Fashion-IQ and CIRCO with ViT-B/32 to provide an in-depth analysis of WISER, including the effectiveness of its core components, compatibility with various modules, hyperparameter sensitivity, and computational efficiency.

Effectiveness of Core Components. Table 3 presents the

Table 1. **Comparison with the state-of-the-art methods on the test sets of CIRCO and CIRR.** The best results are in **bold**, while the second-best results are underlined. ‘-’ indicates results not reported in the original paper. Our method is highlighted in grey.

Backbone	Method	Training-free	CIRCO				CIRR							
			mAP@k				Recall@k				Recall _{sub} @k			
			k = 5	k = 10	k = 25	k = 50	k = 1	k = 5	k = 10	k = 50	k = 1	k = 2	k = 3	
ViT-B/32	PALAVRA (ECCV'22)	✗	4.61	5.32	6.33	6.80	16.62	43.49	58.51	83.95	41.61	65.30	80.94	
	SEARLE (ICCV'23)	✗	9.35	9.94	11.13	11.84	24.00	53.42	66.82	89.78	54.89	76.60	88.19	
	CIReVL (ICLR'24)	✓	14.94	15.42	17.00	17.82	23.94	52.51	66.00	86.95	60.17	80.05	90.19	
	LDRE (SIGIR'24)	✓	17.96	18.32	20.21	21.11	25.69	55.13	69.04	89.90	60.53	80.65	90.70	
	OSrCIR (CVPR'25)	✓	18.04	19.17	20.94	21.85	25.42	54.54	68.19	-	62.31	80.86	91.13	
	AutoCIR (KDD'25)	✓	18.82	19.41	21.38	22.32	30.53	59.42	72.19	<u>91.47</u>	65.11	84.02	92.70	
	CoTMR (ICCV'25)	✓	<u>22.23</u>	<u>22.78</u>	<u>24.68</u>	<u>25.74</u>	<u>31.50</u>	<u>60.80</u>	<u>73.04</u>	91.06	<u>66.61</u>	<u>84.50</u>	<u>92.55</u>	
	WISER (Ours)	✓	32.23	33.18	34.82	35.35	49.45	76.55	85.21	93.81	77.30	88.63	92.27	
ViT-L/14	SEARLE (ICCV'23)	✗	11.68	12.73	14.33	15.12	24.24	52.48	66.29	88.84	53.76	75.01	88.19	
	LinCIR (CVPR'24)	✗	12.59	13.58	15.00	15.85	25.04	53.25	66.68	-	57.11	77.37	88.89	
	MOA (SIGIR'25)	✗	15.30	17.10	18.50	19.30	27.10	56.50	69.20	90.00	-	-	-	
	HIT (ICCV'25)	✗	15.50	16.70	18.90	19.90	27.90	57.60	70.50	90.40	-	-	-	
	CIReVL (ICLR'24)	✓	18.57	19.01	20.89	21.80	24.55	52.31	64.92	86.34	59.54	79.88	89.69	
	LDRE (SIGIR'24)	✓	23.35	24.03	26.44	27.50	26.53	55.57	67.54	88.50	60.43	80.31	89.90	
	IP-CIR (CVPR'25)	✓	26.43	27.41	29.87	31.07	29.76	58.82	71.21	90.41	62.48	81.64	90.89	
	OSrCIR (CVPR'25)	✓	23.87	25.33	27.84	28.97	29.45	57.68	69.86	-	62.12	81.92	91.10	
	AutoCIR (KDD'25)	✓	24.05	25.14	27.35	28.36	31.81	61.95	73.86	92.07	67.21	84.89	<u>93.13</u>	
	CoTMR (ICCV'25)	✓	<u>27.61</u>	<u>28.22</u>	<u>30.61</u>	<u>31.70</u>	<u>35.02</u>	<u>64.75</u>	<u>76.18</u>	<u>92.51</u>	<u>69.39</u>	<u>85.75</u>	93.33	
	WISER (Ours)	✓	35.10	36.30	38.46	39.15	49.23	76.72	85.11	94.17	77.81	88.89	92.77	
ViT-G/14	LinCIR (CVPR'24)	✗	19.71	21.01	23.13	24.18	35.25	64.72	76.05	-	63.35	82.22	91.98	
	CIReVL (ICLR'24)	✓	26.77	27.59	29.96	31.03	34.65	64.29	75.06	91.66	67.95	84.87	93.21	
	LDRE (SIGIR'24)	✓	31.12	32.24	34.95	36.03	36.15	66.39	77.25	93.95	68.82	85.66	93.76	
	IP-CIR (CVPR'25)	✓	32.75	34.26	36.86	38.03	39.25	70.07	80.00	94.89	69.95	86.87	94.22	
	OSrCIR (CVPR'25)	✓	30.47	31.14	35.03	36.59	37.26	67.25	77.33	-	69.22	85.28	93.55	
	CoTMR (ICCV'25)	✓	<u>32.23</u>	<u>32.72</u>	<u>35.60</u>	<u>36.83</u>	<u>36.36</u>	<u>67.52</u>	<u>77.82</u>	<u>93.99</u>	<u>71.19</u>	<u>86.34</u>	<u>93.87</u>	
	WISER (Ours)	✓	36.53	38.14	40.46	41.26	49.54	77.40	85.76	94.17	78.10	89.06	92.68	

ablation results for the core components of our method. We observe the following key findings: (1) Relying solely on either T2I or I2I yields limited performance, with I2I being even weaker than T2I, primarily due to its difficulty in accurately interpreting semantic modifications. These results confirm *the inherent limitations of the single retrieval paradigm under diverse CIR demands*. (2) Simply merging the dual-path with fixed weights at the similarity level (‘AVG’ row) can even degrade performance compared to single-path retrieval. We further compare various fixed fusion strategies with WISER (see Figure 5 in supplementary materials). The results show that WISER significantly outperforms all fixed-weight combinations, highlighting *the limitations of manual weight tuning* and verifying the effectiveness of our Adaptive Fusion in leveraging complementary strengths. (3) Introducing Deeper Thinking on top of Adaptive Fusion leads to further consistent improvements. Moreover, applying Deeper Thinking to single-path baselines also helps. These results indicate that *iterative analysis effectively corrects retrieval errors*. (4) Notably, T2I (using BAGEL) performs comparably to the T2I-

based baseline CIReVL (e.g., 28.59% vs. 28.27% R@10 on Fashion-IQ), indicating no decisive advantage from the editor alone. Our full framework achieves substantial improvements, nearly doubling the single-path performance, which verifies that the gains are attributable to our novel design **rather than merely relying on more powerful models**.

Compatibility with Various Modules. We analyze the effects of different editor, verifier, and refiner modules in Table 4. (1) Refiner: Different LLMs achieve comparable performance, indicating that WISER is robust to the choice of the specific LLM for refinement. (2) Verifier: Performance generally improves with model scale, though the 32B variant shows slight degradation, probably due to overthinking. Despite this, all verifier configurations achieve high performance, surpassing the best baseline. (3) Editor: While performance varies across editing models, our framework maintains satisfactory results. These findings demonstrate WISER’s plug-and-play nature, effectively working with various off-the-shelf models.

Parameter Sensitivity of Threshold and Iteration. We analyze the impact of the confidence threshold τ in Deeper

Table 2. **Comparison with the state-of-the-art methods on the validation set of Fashion-IQ.** The best results are in **bold**, while the second-best results are underlined. Our method is highlighted in grey.

Backbone	Method	Training-free	Shirt		Dress		Toptee		Avg.	
			R@10	R@50	R@10	R@50	R@10	R@50	R@10	R@50
ViT-B/32	PALAVRA (ECCV'22)	✗	21.49	37.05	17.25	35.94	20.55	38.76	19.76	37.25
	SEARLE (ICCV'23)	✗	24.44	41.61	18.54	39.51	25.70	46.46	22.89	42.53
	CIReVL (ICLR'24)	✓	28.36	47.84	25.29	46.36	31.21	53.85	28.29	49.35
	LDRE (SIGIR'24)	✓	27.38	46.27	19.97	41.84	27.07	48.78	24.81	45.63
	OSrCIR (CVPR'25)	✓	31.16	51.13	29.35	50.37	36.51	58.71	32.34	53.40
	AutoCIR (KDD'25)	✓	32.43	51.67	26.52	46.36	33.96	56.09	30.97	51.37
	CoTMR (ICCV'25)	✓	<u>33.42</u>	<u>53.93</u>	<u>31.09</u>	<u>54.54</u>	<u>38.40</u>	<u>61.14</u>	<u>34.30</u>	<u>56.54</u>
	WISER (Ours)	✓	41.51	56.92	37.48	56.97	46.97	62.32	41.99	58.74
ViT-L/14	SEARLE (ICCV'23)	✗	26.89	45.58	20.48	43.13	29.32	49.97	25.56	46.23
	LinCIR (CVPR'24)	✗	29.10	46.81	20.92	42.44	28.81	50.18	26.28	46.49
	MOA (SIGIR'25)	✗	31.90	50.70	25.20	48.50	33.20	54.80	30.10	51.30
	HIT (ICCV'25)	✗	32.40	51.20	25.60	47.10	32.80	54.70	30.30	51.00
	CIReVL (ICLR'24)	✓	29.49	47.40	24.79	44.76	31.36	53.65	28.55	48.57
	LDRE (SIGIR'24)	✓	31.04	51.22	22.93	46.76	31.57	53.64	28.51	50.54
	OSrCIR (CVPR'25)	✓	33.17	52.03	29.70	51.81	36.92	59.27	33.26	54.37
	AutoCIR (KDD'25)	✓	34.00	53.43	24.94	45.81	33.10	55.58	30.68	51.60
	CoTMR (ICCV'25)	✓	<u>35.43</u>	<u>54.91</u>	<u>31.18</u>	<u>55.04</u>	<u>38.55</u>	<u>61.33</u>	<u>35.05</u>	<u>57.09</u>
	WISER (Ours)	✓	43.13	57.85	38.42	55.92	45.39	61.50	42.17	58.51
ViT-G/14	LinCIR (CVPR'24)	✗	46.76	65.11	38.08	60.88	50.48	71.09	45.11	65.69
	CIReVL (ICLR'24)	✓	33.71	51.42	27.07	49.53	35.80	56.14	32.19	52.36
	LDRE (SIGIR'24)	✓	35.94	58.58	26.11	51.12	35.42	56.67	32.49	55.46
	OSrCIR (CVPR'25)	✓	38.65	54.71	33.02	54.78	41.04	61.83	37.57	57.11
	AutoCIR (KDD'25)	✓	36.36	55.84	26.18	47.69	37.28	60.38	33.27	54.63
	CoTMR (ICCV'25)	✓	38.32	<u>62.24</u>	34.51	57.36	41.90	64.30	38.25	61.32
	WISER (Ours)	✓	<u>45.98</u>	61.58	38.57	<u>58.35</u>	<u>49.21</u>	<u>66.96</u>	<u>44.59</u>	<u>62.30</u>

Table 3. **Ablation study on the core components of WISER.** “AVG” denotes average and “ADA” denotes Adaptive Fusion.

Wider Search		Deeper Thinking	Fusion	Fashion-IQ-Avg		CIRCO			
T2I	I2I			R@10	R@50	mAP@5	mAP@10	mAP@25	mAP@50
-	✓	-	-	22.65	38.84	7.00	7.46	8.40	8.95
-	✓	✓	-	23.58	40.10	7.57	8.05	9.04	9.62
✓	-	-	-	28.59	49.18	17.28	17.94	19.64	20.51
✓	-	✓	-	29.22	49.94	17.64	18.30	20.01	20.94
✓	✓	-	AVG	33.40	52.92	13.53	14.30	15.96	16.77
✓	✓	-	ADA	40.83	57.86	31.32	32.08	33.72	34.24
✓	✓	✓	ADA	41.99	58.74	32.23	33.18	34.82	35.35

Thinking (Figure 4, left). As τ increases from 0.4 to 0.8, mAP first rises then slightly declines, indicating that overly low thresholds miss improvement opportunities while overly high ones trigger unnecessary refinement. Performance remains consistently strong when τ lies between 0.5 and 0.7, allowing flexible trade-offs between efficiency and effectiveness. We further examine refinement iterations N (Figure 4, right). All metrics show rapid improvement in the first iteration, with only marginal gains in subsequent rounds. This demonstrates that one iteration generally achieves stable performance, while additional iterations

Table 4. **Ablation study on the choice of editor, verifier, and refiner models on the CIRCO dataset, evaluated with mAP@k.**

Editor	Verifier	Refiner	k=5	k=10	k=25	k=50
Bagel [18]	Qwen2.5-VL-7B [3]	Qwen-Turbo [2]	32.80	33.56	35.21	35.79
		GPT-3.5-Turbo [7]	32.57	33.49	35.13	35.63
		GPT-4o-Mini [7]	32.21	33.06	34.74	35.28
		GPT-4o [1]	32.23	33.18	34.82	35.35
Bagel [18]	Qwen2.5-VL-7B [3]	Qwen2-VL-7B [53]	25.50	26.26	28.41	29.12
		Qwen2.5-VL-3B [3]	27.50	28.45	30.16	30.79
		Qwen2.5-VL-7B [3]	32.23	33.18	34.82	35.35
		Qwen2.5-VL-32B [3]	31.69	32.26	34.22	34.70
GPT4o + OmniGen2 [56] GPT4o + Step1X-Edit [37] GPT4o + Bagel-Edit [18] Bagel [18]	Qwen2.5-VL-7B [3]	GPT-4o [1]	31.18	32.16	33.82	34.34
			31.91	33.02	34.92	35.46
			32.21	33.25	34.99	35.51
			32.23	33.18	34.82	35.35

yield diminishing returns at higher computational cost.

Efficiency Analysis. WISER’s computational overhead is justified by substantial performance gains, as it requires only 0.5 GPU hours per 1% of improvement. Moreover, its plug-and-play architecture enables a flexible trade-off between efficiency and effectiveness, since any component can be replaced with a faster variant. As the most expensive component, Deeper Thinking is triggered only for low-confidence cases, which are determined by the threshold τ . For reasonable values of τ , the refinement rate remains be-

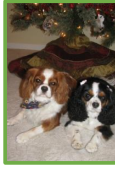
	Multimodal Query	Edited Image	Edited Caption	Suggestions	I2I (Top-1)	T2I (Top-1)	WISER (Top-1)
(a) Fashion-IQ	 has more black designs and has a paint pattern.		The woman is wearing a dress with a floral print, featuring more black designs and a paint pattern.	Good retrieval, no more loops needed.			
(b) CIRR	 two dogs of the same breed on the floor.		Two black and white dogs of the same breed on the floor.	Good retrieval, no more loops needed.			
(c) CIRCO	 has a dog of a different breed and shows a jolly roger.		A dog wearing a different breed hat is in a bike basket. A beagle wearing a red hat is sitting in a bike basket, with a jolly roger flag clearly displayed.	I2I Change dog breed distinctly; add jolly roger symbol visibly on dog's hat or basket. T2I Specify dog breed change and inclusion of a jolly roger flag clearly in the description.			

Figure 3. **Qualitative results on (a) Fashion-IQ, (b) CIRR, and (c) CIRCO datasets.** Red indicates wrong, green represents correct and the gray arrow points to refined results.

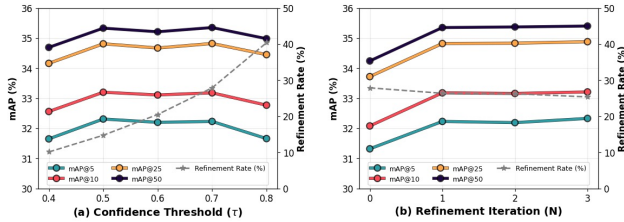


Figure 4. **Sensitivity analysis on confidence threshold τ and refinement iteration N on CIRCO.**

low 30% (see Figure 4), allowing direct retrieval for most queries. Notably, WISER already surpasses prior work in a single retrieval round. Therefore, users can flexibly adjust τ and N based on practical needs.

4.4. Qualitative Results

Figure 3 presents qualitative examples from WISER across three datasets. Each case shows the composed query, edited image, edited caption, refinement suggestions, and top-1 results from T2I-only, I2I-only, and WISER. In Figure 3(a), the query requires preserving the reference’s style while adding pattern changes. T2I captures semantic changes but loses global style cues, while I2I maintains appearance but misunderstands target patterns. WISER successfully reranks the target image to top-1 by prioritizing candidates with higher overall consistency. Figure 3(b) involves fine-grained breed identification and background replacement. T2I over-abstracts breed details, while I2I struggles with

precise retrieval. By jointly verifying breed cues and background changes, WISER dynamically balances visual fidelity and semantic precision to achieve top-1 retrieval. Figure 3(c) demonstrates handling ambiguous modifications. Initial T2I and I2I both fail—T2I misinterprets key intent while I2I misplaces visual elements. As a result, WISER activates Deeper Thinking to refine both paths: enhancing semantic recovery for T2I and correcting visual placement for I2I. In the subsequent retrieval, the improved edited image and caption enable correct top-1 identification, showcasing WISER’s robustness through self-reflection. More details and failure cases are provided in Sec. 7 of the supplementary material.

5. Conclusion

In this work, we introduced WISER, a novel training-free framework for ZS-CIR. To address the diverse nature of real-world user intents, WISER leverages the complementary strengths of both Text-to-Image retrieval (T2I) and Image-to-Image retrieval (I2I) paradigms through a proposed “retrieve–verify–refine” pipeline. Extensive experiments across multiple benchmarks demonstrate that WISER achieves new state-of-the-art performance, significantly outperforming previous training-free methods and even surpassing many training methods. This remarkable improvement confirms WISER’s strong superiority and generalization capability across diverse modification intents.

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